

A STUDY OF THE REDUCTION OF TRIS-OXALATO Fe(III) FROM AQUEOUS 1 M OXALATE SOLUTION AT A POLYCRYSTALLINE GOLD ELECTRODE

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ABSTRACT

The reduction of the Fe(III)-oxalate complex at a polycrystalline gold electrode from a solution of 1 M $K_2C_2O_4 + 0.05$ M $H_2C_2O_4$ has been studied by dc current and impedance measurements. The results indicate a multi-step mechanism of the CEC type, i.e. the electron transfer is preceded and followed by a *heterogeneous* chemical reaction. An EC mechanism (preceding reaction not present or fast) can also fit the experiments if the individual transfer coefficient of the electron-transfer step is allowed to be 0.4 instead of 0.5. The conclusions are compared with earlier results obtained at the dropping mercury electrode.

(1) INTRODUCTION

Electrode reactions involving the transfer of a single electron between two solution-soluble redox components are usually expected to obey a relatively simple rate law describing the potential dependence of the rate of charge transfer. That is, the potential-dependent forward rate constant of reduction, k_f , is expressed in terms of the potential-independent kinetic parameters k_{sh}^t and α by

$$k_f \exp[(z_O F/RT) \phi_2] = k_{sh}^t \exp\left[(-\alpha F/RT)(E - \phi_2 - E^\circ + \phi_2^{E=E^\circ})\right] \quad (1)$$

where ϕ_2 is the potential of the outer Helmholtz plane (OHP), E° the formal standard potential and z_O the charge on the oxidized species in the solution. Obviously in the notation of eqn. (1) the standard heterogeneous rate constant k_{sh}^t is the value of the left-hand side at $E = E^\circ$. Concordance with eqn. (1) has indeed been observed for a number of redox couples of the type mentioned [1–5], and in addition the value of the transfer coefficient α is found to be close to 0.5, thus supporting theoretical views on the essentials of charge transfer, for example the Marcus theory [6].

The tris-oxalato Fe(III)/tris-oxalato Fe(II) electrode reaction at the mercury electrode appears to be an exceptional case, especially concerning the latter aspect, the cathodic transfer coefficient being frequently reported to have an extreme value

between 0.8 and 0.9 [7–11], and recently in our laboratory using the demodulation method we found α to be even closer to unity [12]. The phenomenon was discussed carefully combining results of the demodulation method with those of the impedance method. The conclusion from this work was that the kinetics of this system were not to be described in terms of the Butler–Volmer rate law, as in eqn. (1), but rather pointed to the existence of a two-step mechanism, consisting of the electron-transfer step followed by a chemical step (EC mechanism). Moreover, it was shown that the interpretation of the impedance results alone was complicated by the existence of slight adsorption of both redox components.

Studies of the iron-oxalate reaction at electrode materials other than mercury remained thus far restricted to the determination of the apparent standard heterogeneous rate constant, k_{sh}^a , at platinum, silver and gold by Randles and Somerton [13], by means of the impedance method in the way proposed by Randles [14]. It is remarkable that at these metals the k_{sh}^a values are ten to a hundred times lower than at mercury: respectively 1×10^{-3} (Ag), 5×10^{-3} (Au), 9×10^{-3} (Pt) [13] and 1.5 (Hg) [12] cm s^{-1} . It will be interesting therefore to investigate the potential-dependent rate constant at one of the other metals. If the two-step mechanism applies, this must show up quite clearly in the results.

Recently, we have focused our attention to the development of procedures to perform impedance measurements at gold electrodes and to analyse the resulting data in such a way that the reliability of the conclusions can be guaranteed to a large extent [15]. In particular, the interference introduced by the so-called constant phase element (CPE), representing the impedance of the electrical double-layer, was considered and shown to be accessible for elimination also in the case of the iron-oxalate system.

In this paper we wish to report and to discuss the results of a complete impedance analysis of the iron-oxalate–gold system, which appeared to be possible in quite a wide potential range, viz. from -0.100 to -0.480 V vs. SCE. It will be shown that, regardless of the Frumkin effect, the $\ln k_f$ vs. E curve is appreciably curved, supporting the idea of a multi-step mechanism.

(II) EXPERIMENTAL

The reduction of the Fe(III)-oxalate complex was studied at a flat polycrystalline gold electrode of 99.999% purity and a geometrical area of 0.071 cm^2 in contact with a solution containing $1 \text{ M K}_2\text{C}_2\text{O}_4 + 0.05 \text{ M H}_2\text{C}_2\text{O}_4$ and $2 \times 10^{-3} \text{ M Fe}_2(\text{SO}_4)_3$. The contact was achieved by means of the pendant meniscus method, i.e. by dipping the electrode on the solution surface so that the cylindrical walls were just not wetted. The solution was prepared by dissolving oxalic acid and potassium carbonate (both Merck suprapur) in freshly twice-distilled water. The solution was deaerated by means of purified argon and kept in the dark.

The measuring technique was basically the same as that described earlier [15a].

The dc-current or the ac-impedance was measured 4 s after the application of a potential step from a initial potential into the faradaic region. After each measure-

ment the potential was swept from the initial potential (0.00 V vs. SCE) to a more positive potential (0.25 V vs. SCE) in order to eliminate the concentration gradients of $\text{Fe(III)(C}_2\text{O}_4)_3^{3+}$ and $\text{Fe(II)(C}_2\text{O}_4)_3^{4-}$.

In oxalate solutions it is not possible to clean the electrode surface electrochemically by sweeping into the potential region where gold is oxidized since then the gold dissolves to a considerable extent and oxalate is probably oxidized as well. The positive end of the sweep was therefore kept below the onset of an anodic current.

The cell construction was as described in ref 15a. In the actual measurements, the potential was stepped to potentials ranging from -0.100 to -0.480 V vs. SCE. Impedance measurements were made at twelve fixed frequencies between 80 Hz and 10 kHz. Both dc and ac measurements were made as a function of dc potential, in the region $-0.05 > E > -0.5$ V vs. the saturated calomel electrode (SCE). The cell was thermostated at 25°C .

(III) DATA ANALYSIS

The reduction of tris-oxalato Fe(III) is expected to obey a rate equation which is first order in c_{O} and c_{R} :

$$-J_{\text{F}} = Fk_{\text{f}}[c_{\text{O}} - c_{\text{R}} \exp(\phi)] \quad (2)$$

with

$$\phi = (F/RT)(E - E^\circ) \quad (3)$$

where c_{O} and c_{R} are the surface concentrations and k_{f} is the potential-dependent rate constant. The mass transport of reactant and product will proceed via semi-infinite linear diffusion.

The dc faradaic current flowing after application of a potential step will have a current density given by [16]:

$$-J_{\text{F,dc}} = Fk_{\text{f}}c_{\text{O}}^* \exp(\lambda^2 t) \operatorname{erfc}(\lambda t^{1/2}) \quad (4)$$

with

$$\lambda = k_{\text{f}}[D_{\text{O}}^{-1/2} + D_{\text{R}}^{-1/2} \exp(\phi)] \quad (5)$$

where c_{O}^* is the bulk concentration of the oxidized species and D_{O} and D_{R} are the diffusion coefficients. The limiting current density is found for $\exp(\phi) \rightarrow 0$ and $k_{\text{f}} \rightarrow \infty$:

$$-J_{\text{F,l}} = Fc_{\text{O}}^* D_{\text{O}}^{1/2} / (\pi t)^{1/2} \quad (6)$$

In view of the literature value of the apparent standard rate constant, $k_{\text{sh}} = 5 \times 10^{-3} \text{ cm s}^{-1}$ [13], it can be expected that the dc voltammogram will approach closely the limiting behaviour of a dc reversible system, corresponding to the relationship

$$J_{\text{F,l}}/J_{\text{F,dc}} = 1 + (D_{\text{O}}/D_{\text{R}})^{1/2} \exp(\phi) \quad (7)$$

Consequently the dc voltammogram can be employed to obtain the values of D_O and $E_{1/2}^r$, the reversible half-wave potential, if necessary by iteration, when eqn. (7) is not rigorously obeyed. As in the present case we can assume $D_R = D_O$, this immediately provides the standard potential E° .

The impedance analysis will, as argued before [15], most advantageously be performed by means of the quantity

$$Y_E' - Y_E'' \operatorname{tg}(\alpha_r \pi / 2) = R_{ct}^{-1} \frac{p^2 + p[1 - \operatorname{tg}(\alpha_r \pi / 2)]}{p^2 + 2p + 2} \quad (8)$$

where R_{ct} is the charge-transfer resistance:

$$R_{ct}^{-1} = \frac{F^2}{RT} \frac{k_f c_O^*}{1 + (D_R/D_O)^{1/2} \exp(-\phi)} \times \left\{ 1 + \left[\alpha + \alpha \left(\frac{D_R}{D_O} \right)^{1/2} \exp(-\phi) - 1 \right] \exp(\lambda^2 t) \operatorname{erfc}(\lambda t^{1/2}) \right\} \quad (9)$$

and p is the irreversibility quotient:

$$p = \omega^{1/2} p' = (2\omega D_O)^{1/2} k_f^{-1} \left[1 + \left(\frac{D_O}{D_R} \right)^{1/2} \exp(\phi) \right]^{-1} \quad (10)$$

In eqn. (9), α is the operational transfer coefficient, defined as

$$\alpha = -d \ln k_f / d\phi \quad (11)$$

The left-hand side of eqn. (8) is a combination of the real component Y_E' and the imaginary component Y_E'' of the interfacial admittance, obtained from impedance data after subtraction of the ohmic resistance R_Ω as usual [17], and the rotation parameter α_r of the constant phase element [15]. As discussed in ref. 15, iterative adjustments of R_Ω and α_r may be required in order to arrive, via frequency spectrum analysis, at values for R_{ct} and p' that are consistent in giving the same k_f value or, to put it in another way, a value for the Warburg coefficient $\sigma = R_{ct}/p'$ that is in agreement with its theoretical value. As long as such adjustments are within the limits of accuracy, this is an acceptable procedure.

Direction determination of k_f from R_{ct}^{-1} at a certain potential is not possible in cases where in eqn. (9) the term between brackets differs significantly from unity. In addition, it can occur that the p' value is obtained with a large uncertainty. In such cases we found it useful to fit our data pertaining to a small potential region (say 60 mV wide) to an approximate exponential expression for k_f of the form

$$k_f = A_1 \exp(-A_2 \phi) \quad (12)$$

where A_2 bears some resemblance to α in eqn. (11). The fit can be applied either to R_{ct}^{-1} , cf. eqn. (9), or to the complete admittance function, cf. eqn. (8), at all the frequencies of the experiment. Applying this procedure to subsequent partly overlapping potential ranges, a $\ln k_f$ vs. potential curve could be constructed from the linear segments obtained following eqn. (12).

The final results for k_f and α were inserted into eqns. (9), (10) and (5) for a re-calculation of R_{ct}^{-1} and p' , to be used in the re-calculation of the quantity $Y'_E - Y''_E \operatorname{tg}(\alpha_r \pi/2)$ in eqn. (8). As an additional check, the value of the CPE parameter Q^{-1} was calculated from the imaginary admittance component

$$Y''_E = R_{ct}^{-1} \frac{p}{p^2 + 2p + 2} + \omega^{-1} Q^{-1} \cos(\alpha_r \pi/2) \quad (13)$$

If the complete analysis was correct, Q^{-1} should be found to be independent of frequency.

(IV) RESULTS AND DISCUSSION

(IV.1) The dc voltammogram

From the limiting current, the diffusion coefficient D_O was determined: $D_O = 5.6 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$. This value is about 8% higher than recently obtained at the dropping mercury electrode [12]. However, it is questionable whether this discrepancy (4% in $D_O^{1/2}$) is significant, in view of the possible inadequacy of the diffusion models in the case of both electrodes (linear diffusion for the gold electrode and the diffusion layer approximation for the DME). Moreover, the surface area of the gold electrode could be estimated with an inaccuracy of a few percent.

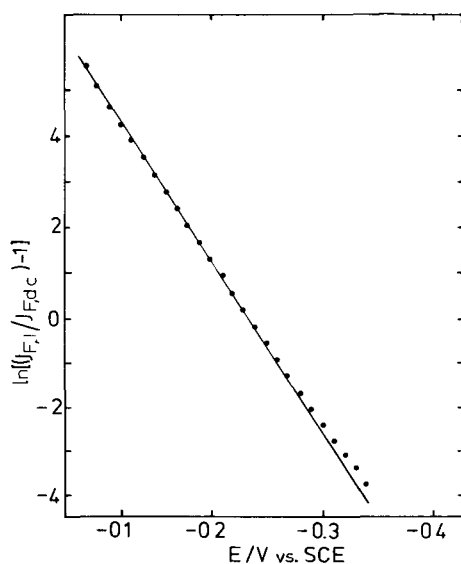


Fig. 1. Plot of $\ln[(J_{F1}/J_{Fdc}) - 1]$ vs. potential for the reduction of Fe(III)-oxalate from 1 M $K_2C_2O_4 + 0.05$ M $H_2C_2O_4$ solution. (●) Measured; (—) calculated from eqns. (4), (5) and (14) using the parameter values reported in text.

In Fig. 1 the voltammogram is analysed by plotting $\ln[(J_{F,L}/J_{F,dc}) - 1]$ against potential. The slope of this plot is 39.5 V^{-1} , which is close to the theoretical "reversible" slope of 38.94 V^{-1} . It is therefore concluded that no information concerning the rate constant can be obtained from these dc experiments.

The logarithmic plot gives the reversible half-wave potential and if $D_R = D_O$, the standard potential: $E^\circ = -0.232 \pm 0.002 \text{ V}$ vs. SCE, in very good agreement with $E^\circ = -0.231 \text{ V}$ vs. SCE, found at the dropping mercury electrode [12].

(IV.2) The admittance measurements

In Fig. 2 we plotted the quantity $Y'_E - Y''_E \text{tg}(\alpha_r \pi/2)$ against potential at two frequencies, viz. 120 and 780 Hz. It can be seen that the frequency dispersion is relatively small, indicating that the electrode reaction is almost ac irreversible. Indeed, when we analysed all the data in the way described in ref. 15, the values obtained for R_{ct}^{-1} were quite close to the upper curve in Fig. 2. The value of α_r involved in the analysis could be adjusted with a precision of 0.005 and was found to be potential-dependent: α_r decreases gradually from 0.05 at $E = -0.048$ to 0.025 at $E = -0.479 \text{ V}$ vs. SCE, i.e. a change of the rotation angle of the CPE of 4.5 to 2.25° . In the separate analysis of Y''_E , the best values of α_r to give a frequency-independent result for Q^{-1} were systematically somewhat higher, varying from 0.06 (i.e. 5.4°) at $E = -0.048$ to 0.3 (i.e. 2.7°) at $E = -0.479 \text{ V}$ vs. SCE.

It can also be seen in Fig. 2 that the admittance data give valuable information in a large potential range, from ca. $+0.12$ to ca. -0.24 V with respect to the standard potential. Applying the procedures described in Section (III), we determined the complete set of R_{ct}^{-1} and p' values and the corresponding values of the forward rate constant. The solid curves in Fig. 2 represent the values of $Y'_E - Y''_E \text{tg}(\alpha_r \pi/2)$ re-calculated from these results. The consistency of the analysis, in the sense discussed before [15], is clearly demonstrated in Fig. 3, where the solid curve represents $\ln k_f$ vs. E as obtained by the "direct" fitting of $Y'_E - Y''_E \text{tg}(\alpha_r \pi/2)$ values

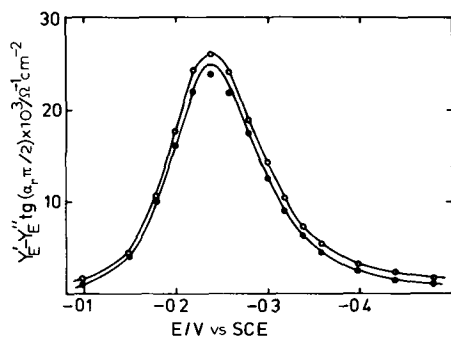


Fig. 2 Plot of $Y'_E - Y''_E \text{tg}(\alpha_r \pi/2)$ vs. potential at the frequencies 120 Hz (●) and 780 Hz (○). (—) Curves calculated from eqns (8)–(11) using values for R_{ct}^{-1} and p' obtained from frequency analysis.

in small potential ranges using the empirical eqn. (12), and the open and closed circles represent the $\ln k_f$ values calculated from R_{ct}^{-1} and p' , respectively. These results were obtained after assigning the value $E^\circ = -0.232$ V vs. SCE to the standard potential and the value $D_O = 4.8 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ to the diffusion coefficient of tris-oxalato Fe(III). The latter value is somewhat lower than the value determined from the dc voltammogram. It should be noted, however, that on the ac time scale the system is almost entirely controlled by charge transfer, so that the deviation might well be due to a slight systematic error in the admittance analysis.

(IV.3) The reaction mechanism

The logarithm of the forward rate constant is clearly a non-linear function of the dc potential, with limiting slopes corresponding to values for the operational transfer coefficient $\alpha \approx 0.8$ at $E \gg E^\circ$ and $\alpha \approx 0.13$ at $E \ll E^\circ$. In view of our recent discussions about the possibility of multi-step mechanisms [18,19] and in view of the conclusion drawn for the reduction of tris-oxalato Fe(III) at the mercury electrode [12], it is also logical to consider the possibility of such a mechanism here. The steep change of α with potential suggests that k_f could possibly be adapted to a so-called CEC mechanism, i.e. the electron transfer is both *preceded* and *followed* by a "heterogeneous chemical reaction step", the rate constant of which is independent of potential. If it is assumed that the rate constant of the electron-transfer step is related to the potential simply by $k_1 = k_1^\circ \exp(-\alpha_1\phi)$, with α_1 close to 0.5, the overall rate constant will be given by [18, 19]:

$$\frac{1}{k_f} = \frac{1}{k_{c,1}} + \frac{\exp(\alpha_1\phi)}{k_{s,1}} + \frac{\exp(\phi)}{k_{c,-2}} \quad (14)$$

where $k_{c,1}$ is the forward rate constant of the preceding reaction, $k_{s,1} = K_{c,1}^{-1}k_1^\circ$ and $k_{c,-2}$ is the backward rate constant of the following reaction. A quite acceptable fit

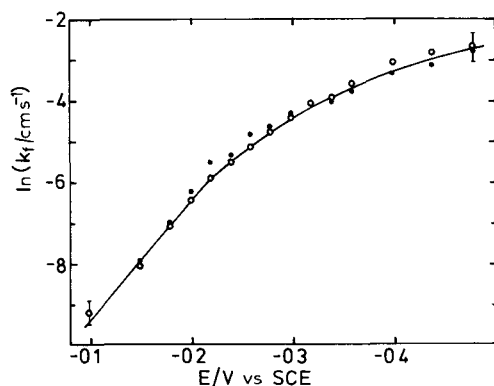


Fig. 3. Plot of $\ln k_f$ vs. potential. (—) Obtained from fit using eqn. (12) (see text), (O) obtained from R_{ct} , eqn (9); (●) obtained from p' , eqn (10).

to eqn. (14) is obtained with $\alpha_1 = 0.5$ and $k_{c,1} = 0.05$, $k_{s,1} = 0.005$, $k_{c,-2} = 0.015 \text{ cm s}^{-1}$. An excellent fit is obtained if a slight potential dependence of the rate of the first step is admitted, viz.: $k_{c,1} = 0.045 \exp[-0.03\phi]$, $k_{s,1} = 0.005$, $k_{c,-2} = 0.015 \text{ cm s}^{-1}$.

However, in view of the high negative charge of the electroactive species and the large potential range involved, it should be envisaged that the non-linear potential dependence of $\ln k_f$ is (partially) due to the variation in the double-layer structure. In order to estimate this effect, the Frumkin correction should be applied in a way analogous to eqn. (1), i.e. by calculating $\ln k_f + (z_O F/RT)\phi_2$ and plotting this quantity vs. $E - \phi_2$. In the present case $z_O = -3$.

Values of ϕ_2 can be calculated by means of the well-known relation

$$\sigma^M = \pm (2\epsilon\epsilon_0 RT)^{1/2} \left[\sum_i c_i \exp(-z_i F\phi_2/RT) - 1 \right]^{1/2} \quad (15)$$

where σ^M is the charge density on the metal and $\epsilon\epsilon_0$ is the permittivity of the solvent ($6.95 \times 10^{-12} \text{ F cm}^{-1}$ for water at 25°C). The summation is made over all the ionic species with concentration c_i just outside the double-layer and valency z_i . The electroactive species are also included because of their high charge [4]. Furthermore, it is assumed that potassium oxalate is completely dissociated, whereas the degree of dissociation of oxalic acid is calculated from its dissociation constant [20].

The common way to determine σ^M as a function of E is to integrate the double-layer capacity. In our previous paper it was argued that it is unclear which quantity should be considered as the "double-layer capacity" of an electrode/solution interface in the case where the electrical double-layer behaves as a constant phase element. With the present state of knowledge, we feel that the most unambiguous quantity will be the capacity C_0 , which is related to the CPE parameter Q^{-1} by [15]:

$$Q^{-1} = C_0^{1-\alpha_r} R_\Omega^{-\alpha_r} \quad (16)$$

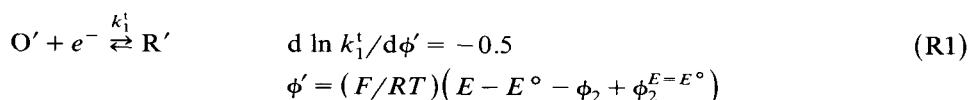
where R_Ω is the ohmic resistance of the solution, in the present case equal to $1.12 \Omega \text{ cm}^2$. Its influence is, in this case, negligible since $R_\Omega^{-\alpha_r} \approx 1$. It is easily seen that C_0 has the dimensions F cm^{-2} and that C_0 is defined as the classical double-layer capacity in the case where the rotation parameter α_r equals zero. An argument supporting the choice of C_0 to represent the quantity $d\sigma^M/dE$ is the fact that its magnitude and its potential dependence are found to be rather similar to the double-layer capacitance C_d of the dropping mercury electrode in a $1 \text{ M K}_2\text{C}_2\text{O}_4 + 0.05 \text{ M H}_2\text{C}_2\text{O}_4$ solution. In Fig. 4 both the quantities C_0 derived from the admittance analysis at the gold electrode and C_d measured directly at the DME [12] are plotted against $E - E_{\text{pzc}}$, where E_{pzc} is the potential of zero charge, being -0.040 V vs. SCE for gold and -0.470 V vs. SCE for mercury. The former value was set equal to the literature value of the pzc of gold in perchlorate solutions [21]. The latter value was determined by integration of the C_d vs. E curve of mercury in $1 \text{ M K}_2\text{C}_2\text{O}_4$ from $E = -1.1 \text{ V}$ vs. SCE, where the charge density was taken to be equal

to that of mercury in 1 M KF, $\sigma^M = 12.9 \times 10^{-6} \text{ C cm}^{-2}$. The capacity curves of 1 M $\text{K}_2\text{C}_2\text{O}_4$ and 1 M KF coincide at $E < -1.0 \text{ V}$ vs. SCE.

After integration of C_0 from $E = -0.04$, eqn. (15) was used to compute the ϕ_2 values, assuming that specific adsorption of oxalate anions at gold is negligible. Next, the curve drawn in Fig. 3 was transformed into a "Frumkin corrected plot" of $\ln k_f^t = \ln k_f - (3F/RT)\phi_2$ vs. $E - \phi_2$. The result is represented in Fig. 5 by the open circles. The graph is still non-linear, with similar limiting slopes as in Fig. 3. The line drawn in Fig. 5 represents a curve calculated by means of eqn. (14), where k_f is replaced by k_f^t , using the following values for the "true" individual rate constants, referred to the potential $E^\circ - \phi_2$ ($E = E^\circ$) = -0.220 V vs. SCE:

$$k_{c,1}^t = 2.0 \quad k_{s,1}^t = 0.035, \quad k_{c,-2}^t = 0.022 \text{ cm s}^{-1} \\ \alpha_1 = 0.5$$

Both "chemical" rate constants can be taken to be independent of potential after this Frumkin correction, and the individual transfer coefficient α_1 equals 0.5 exactly. So, these results indicate the following reaction mechanism:



The value of $k_{c,1}$ is relatively large and consequently the preceding chemical step is rate-determining only in the very negative potential region ($E < -0.4 \text{ V}$ vs. SCE). It should be noted that a good fit can also be obtained with $k_{c,1}^t$ infinite, taking $\alpha_1 = 0.4$ instead of 0.5 and slightly adjusting the values of $k_{s,1}^t$ and $k_{c,-2}^t$. The dashed

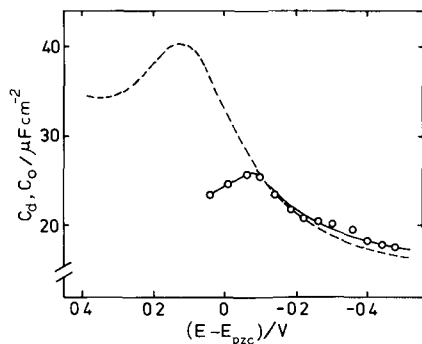


Fig. 4 Comparison of the CPE parameter C_0 (see eqn. 16) for the gold electrode in 1 M $\text{K}_2\text{C}_2\text{O}_4 + 0.05 \text{ M H}_2\text{C}_2\text{O}_4$ (○) with the double-layer capacity C_d of the dropping mercury electrode in the same solution (---). Both quantities are plotted against $E - E_{pzc}$

line in Fig. 5 represents the curve calculated with the following values of the rate parameters:

$$k_{c,1}^t = 1000, \quad k_{s,1}^t = 0.044, \quad k_{c,-2}^t = 0.021 \text{ cm s}^{-1} \\ \alpha_1 = 0.4$$

It is clear that this curve agrees with the experimental results as well.

Finally, it will be interesting to compare these results quantitatively with those of the earlier study [12] of the tris-oxalato iron system at mercury. To this end, in the mercury case also the data should be corrected for the double-layer effect. However, at mercury the faradaic potential region ($-0.1 < E < -0.35$ V vs. SCE) is situated positive from the potential of zero charge, E_{pzc} , and in view of the value of $E_{pzc} = -0.470$ V vs. SCE some specific adsorption of oxalate anions must be expected. To our knowledge, no literature data on the amount of specific oxalate adsorption at these potentials are available. It is therefore impossible to determine precise values of the OHP potential ϕ_2 . From a qualitative comparison of E_{pzc} and the C_d vs. E curve for the oxalate solution (see Fig. 4) with those of other supporting electrolytes, e.g. of 1 M NaClO₄ [22] and 0.1 M KAc [23], we can only surmise that in the potential range of interest ϕ_2 will be close to zero or slightly negative. Therefore in Fig. 5 we reproduced the $\ln k_f$ vs. E curve at mercury without any Frumkin correction. Inevitably the conclusion is that at mercury all individual rate constants are about a hundred times, or more, higher than at gold.

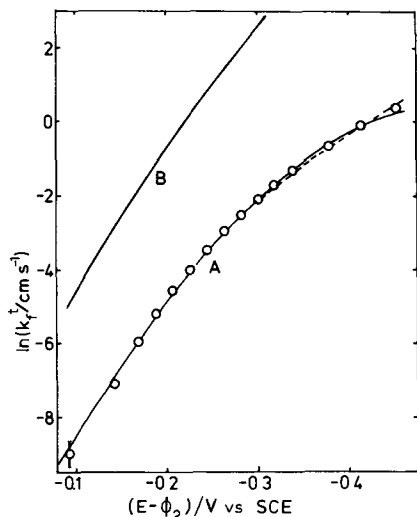


Fig. 5. Potential dependence of the Frumkin-corrected rate constant at gold (A) and of the apparent rate constant at mercury (B) of the reduction of Fe(III)-oxalate from 1 M K₂C₂O₄ + 0.05 M H₂C₂O₄ solution (see text).

(V) CONCLUSION

The most important result from the present work is that, due to the slower reaction rates, the curvature of the $\ln k_f$ vs. E curve at gold is much more pronounced than at mercury. The operational transfer coefficient is found to change from 0.97 at $E - E^\circ = 0.120$ V, via 0.8 at $E = E^\circ$, to ca. 0.2 at $E - E^\circ = -0.240$ V. We feel that this supports the idea that the kinetics of the reaction should be described in terms of a multi-step mechanism, in this case denoted as CEC, rather than on the basis of some reorganizational model that predicts the operational transfer coefficient to be close to 0.5 at $E = E^\circ$ [6].

Generally it is expected that the rates of simple redox reactions do not depend on the nature of the electrode metal. Here it is found most clearly to be the opposite for all steps in the mechanism, i.e. both for the electron transfer and the chemical steps. This is not surprising if one recognizes that we are dealing with heterogeneous processes at interfaces with different properties. At present, for lack of solid identification of the nature of the chemical steps, there is no reason to speculate beyond our earlier suggestion that they could be preceding organizations and following relaxation processes, respectively. Such processes need not necessarily pertain to the solvent, but here also could concern the ligands because the ferric oxalate complex is much stronger than the ferrous oxalate complex, as is reflected by the rather large difference between the stability constants [24].

The number of steps in the mechanism to be assumed to explain the experimental facts relies heavily on the solidity of the idea that the transfer coefficient α_1 of the electron-transfer step should be very close to 0.5, or the idea that it need not be necessarily so. Sound unassailable theory predicting the value of the symmetry factor of a single electron transfer is essential in these analyses.

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